

Vite vs Webpack – Explained Simply

1. Introduction

Modern web applications consist of many files:

- JavaScript
- CSS
- Images
- Components

These need to be optimized for the browser.

Tools like the following help with that:

- Vite
 - Webpack
-

2. What is Webpack?

Webpack is a so-called "module bundler".

That means:

- It takes many files
- Combines them into one bundle (e.g. bundle.js)

Process:

1. Entry point
2. Finds all imports
3. Builds a dependency graph
4. Creates a bundle

Downsides:

- Slow startup time
 - Complex configuration
-

3. What is Vite?

Vite is a modern build tool.

Key difference:

In development, Vite uses:

→ Native ES modules in the browser

This means:

- No full bundling at startup
- Extremely fast dev server

4. Core Difference

Webpack:

- Bundles EVERYTHING before starting

Vite:

- Loads files on demand
- Uses the browser directly

5. Comparison

Feature	Vite	Webpack
Startup time	Very fast	Slow
Setup	Simple	Complex
Dev Server	Instant	Waits for bundle
Build	Rollup	Webpack
Learning curve	Easy	Steep

6. Example Project

Webpack Structure

```
project/  
├─ src/  
│  └─ index.js  
├─ dist/  
└─ webpack.config.js
```

Run:

```
npx webpack
```

Vite Structure

```
project/  
├─ index.html  
├─ main.js
```

Run:

npm run dev

→ Ready instantly

7. Code Example

main.js

```
import { add } from './math.js';  
  
console.log(add(2, 3));
```

math.js

```
export function add(a, b) {  
  return a + b;  
}
```

8. What happens internally?

Vite:

- Dev: no bundling
- Build: uses Rollup

Webpack:

- Dev: bundling
 - Build: bundling
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9. When to use what?

Use Vite for:

- New projects
- Fast development

- Modern apps

Use Webpack for:

- Existing (legacy) projects
 - Special plugins
 - Large enterprise systems
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10. Conclusion

Vite is usually the better choice for new projects:

- faster
- simpler
- more modern

Webpack remains relevant for existing systems.

11. TL;DR

- Vite replaces Webpack in most cases
- No bundling in development
- Uses Rollup for production
- Much better developer experience